Nattanitcha Promboon

PRODUCT DESIGN & PROTOTYPING · CO-OP INTERN

nattanitchafarm@gmail.com
Portfolio

Full-time co-op · 13 Jan – 28 May 2027

EDUCATION

King Mongkut's University of Technology Thonburi (KMUTT)
B.Sc. Media Technology
2023 – present · co-op track
GPA 3.61 / 4.00

Wittayanukulnaree School
Science, Mathematics, Technology & Environment program

SKILLS

Design / Prototyping
Sketching & illustration · UI & motion design · CAD → 3D printing on my own printer · assembly & soldering · sewing & embroidery

Electronics
ESP32 · Arduino · Hall sensors · addressable LEDs · 18650 pack with BMS & USB-C charging

Tools
Figma · Tinkercad · SolidWorks (learning) · Blender · Maya · Photoshop · Illustrator · Creality Print · VS Code · Word · Excel · PowerPoint

AI workflow
I direct AI agents (Claude Code, Codex, Antigravity) using work orders with my exact requirements. A second AI reviews each output. I repeat tasks, compare results and log every step.

AWARDS & ACTIVITIES

1st Place: KMUTT Deep Sea Jam Hackathon 2026 (DeepCore)

Staff: FANTASTECH Media Exhibition 2026, KMUTT

Freelance: commission illustration & handmade dolls (ongoing)

LANGUAGES

Thai: native
English: working proficiency (reading & writing technical documents)

PROFILE

Fourth-year Media Technology student designing and building physical prototypes through concept development, CAD, my own 3D printer, wiring and soldering. I revise designs after failed tests and have led every team project. I handle design and specifications, direct AI to write most of the code, and check the results myself.

SELECTED PROJECTS

METO: small interactive robot for learning block coding — Research → thesis · 2025–present

- I lead a 3-person thesis team comparing beginner block coding with two robots, one showing emotion and one without. The study with first-year KMUTT students is planned, with ethics documents in preparation. Its camera reads facial expressions, and the robot turns to follow the user and mirrors them on its screen.
- "METO Robot" web: I designed its screens and 62-piece avatar. Each account has teacher or student roles set by class. A skill counts only when code runs without error. AI gives hints, not answers.
- Body & mechanism: I chose a trapezoid body so the arms can close in and grip. Five stacked layers assemble without heat-set metal inserts, letting learners build and modify the robot themselves. The Apr 2026 version's printed oval arms with anti-slip bumps lifted a cup. The current tracked version, about 25–30 cm tall, 20–25 cm wide and 30–35 cm deep, runs wirelessly on two rechargeable 18650 cells with a protection board (BMS) and USB-C charging. Car wheels became printed tracks. Two wheels and a caster are planned next.

Magnetic Universe: magnetic LED wall panel — Jan – May 2026 · course project · team of 4

- Light gathers under each magnet pen you place (up to two), and two pools of light blend into one. The 400 × 400 mm panel has 16 Hall sensors (4 × 4). I argued against averaging all 16, so position is computed from the strongest sensor and its neighbours, weighted by field strength.
- I designed and printed threaded rods with twist-on caps, then found the threads were too thin for the slicer to print. The LEDs still showed as dots, so I packed heat-resistant fibre fill into the sensor layer to diffuse the light. This avoided thickening the 6 cm frame, which already weighs 3–4 kg and is mounted at 60°.

DeepCore → DeepSphere: hands-on deep-sea explorer — 2026 – present · 1st Place · team lead

- I chose a physical device over a game, map or dashboard. It features 44 real species backed by 10 peer-reviewed sources, with no invented animals. DeepCore, the hackathon winner, uses an ESP32 controller assembled from real parts to drive a live 3D web scene. I set which sounds we needed, and a teammate sourced them.
- DeepSphere is in progress. The six-screen ring is built, but most screens need re-wiring. The three-layer sphere is still the long-term concept. I set sizes and print rules. I specify support-free printing and thin walls, and use snap-fits because I don't have heat-set inserts. I modelled v1 myself and made the v2 CAD with AI help.

MOFYCH Commission Website: service design — honeykung.github.io/commission-tag

- I designed the whole site in Figma around an "art + web" idea, with hand-drawn clouds, logo and my own animation. Opening alerts use email sign-ups and email-confirmed cancellations. The queue sheet runs on Google Apps Script. The paint matcher covers 1,131 paints from 5 brands. I transcribed most charts, checked the rest against published data, set a $\Delta E < 5$ target and removed fluorescent and metallic paints from 2-paint mix recipes. AI wrote the standard CIEDE2000 matching formulas.

Digital Puzzle Box — prototype

- When the gyro failed, I switched from a flippable box to a flat board. A real hidden key completes the on-screen puzzle.

Smaller builds — self-initiated

- **Minecraft Pixel Lamp**: designed in Tinkercad and printed myself. ESP32 + LEDs. Brightness and current are capped in code (5 V, 1.2 A) so it runs from a phone charger. I specified a flame that steps yellow → orange → red from bottom to top. Size: 10 × 10 × 15.4 cm.
- **svgtostl**: a tool that turns drawings into printable models. I use it for keychains and 3D-printed sewing templates for tracing and cutting fabric. I required sharp corners and accurate curves, rejecting point reduction because it made curves visibly faceted at the scale of 0.2 mm-nozzle printing.
- **Cotton dolls**: made from purchased/public patterns, with my character design sheets, embroidered faces and removable magnetic ears. I make OC-only dolls because purchased patterns bar licensed characters. I am developing my own pattern in Blender.
- **OC Companion** (in design): a palm-sized device (about 4–5 cm per side) with a small AMOLED screen as its whole face and figure-style sculpted hair. It sits on a desk, travels with you and will be made as a keychain. Each device will be made to order for a buyer's own character (OC).
- **Mini paint palette**: 18 wells in a 109 × 75 mm PLA tray (most small palettes have 5–10), with a side channel for test mixing. It is in use.
- **MOFYCH collector cards**: 4 designs × 4 cards, 90 × 50 mm. The artwork is split into layers for holographic foil and matte/smooth finishes, then printed by a factory.